FENGWEI FRIED RICE · STAGE 2 TEST BANK

Mama's Quiz Kitchen

PAPER EDITION

AI Hardware · Networks · Reinforcement Learning · Sensors · Scale

60 questions · 5 kitchen stations · tick, check, collect stars!





How to play on paper

1. Pick a kitchen station (each one is a recipe from our Study Guide). Every station has 5 True/False, 5 multiple-choice and 2 open questions.
2. Tick ☐ your answers. In Word you can click the boxes; on paper, use a pen. "Choose ONE" means one box; "Select ALL" can mean several.
3. Tap "Mama's answer ▸" under a question to jump to its answer, then "◂ Back" to return. No peeking before you answer!
4. Score each station on its scorecard and colour your stars. Then try the game version: Mamas_Quiz_Kitchen.html

True/False	Multiple choice	Open	Stars per station
1 point each	1 point each (all boxes right)	Up to 5 marks each	★★★ 18–20 · ★★ 14–17 · ★ 10–13

Station 1	Station 2	Station 3	Station 4	Station 5	TOTAL
____ / 20 ☆ ☆ ☆	____ / 20 ☆ ☆ ☆	____ / 20 ☆ ☆ ☆	____ / 20 ☆ ☆ ☆	____ / 20 ☆ ☆ ☆	____ / 100 Gold 90+ · Silver 70+ Bronze 50+

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STATION 1

The Kitchen Crew

AI hardware · 5 T/F · 5 multiple choice · 2 open





Mama's hint for Station 1

Count the work first: 1 MAC = 2 FLOPs, and the ridge point tells you if memory or maths is the bottleneck.

Q1 • **TRUE OR FALSE** · §1.1 · Remember

One MAC (multiply-accumulate) counts as two FLOPs.

☐ **TRUE (Tasty!)** ☐ **FALSE (Burnt!)**

Mama's answer ▸

Q2 • **TRUE OR FALSE** · §1.2 · Understand

A TPU is more flexible than a CPU for general programs.

☐ **TRUE (Tasty!)** ☐ **FALSE (Burnt!)**

Mama's answer ▸

Q3 • **TRUE OR FALSE** · §1.3 · Apply

If 90% of a job can be parallelised, 100 cooks give a speed-up of 100×.

☐ **TRUE (Tasty!)** ☐ **FALSE (Burnt!)**

Mama's answer ▸

Q4 • **TRUE OR FALSE** · §1.4 · Understand

At batch size 1, a large layer on a typical accelerator is usually memory-bound.

☐ **TRUE (Tasty!)** ☐ **FALSE (Burnt!)**

Mama's answer ▸

Q5 • **TRUE OR FALSE** · §Lab 1 · Apply · Kitchen Lab

In Kitchen Lab 1 (Race the Crew), the GPU's speed-up over the CPU is LARGER for 256 × 256 matrices than for 4,096 × 4,096 matrices.

☐ **TRUE (Tasty!)** ☐ **FALSE (Burnt!)**

Mama's answer ▸

Q26 • **MULTIPLE CHOICE** · §1.1 · Apply

Choose ONE

Weights [2, −1, 0.5], inputs [1, 4, 6]. What is the neuron output (no bias)?

☐ **A)** 1.0
☐ **B)** 3
☐ **C)** 5
☐ **D)** −1

Mama's answer ▸

Q27 • **MULTIPLE CHOICE** · §1.2 · Understand

Choose ONE

Which chip best fits a voice assistant in a small battery-powered speaker?

☐ **A)** GPU
☐ **B)** Server CPU
☐ **C)** NPU
☐ **D)** TPU pod

Mama's answer ▸

Q28 • **MULTIPLE CHOICE** · §1.2 · Apply

Choose ONE

In a 2 × 2 systolic array with W = [[1, 2], [3, 4]] and x = [5, 6], the output y = W·x is:

☐ **A)** [12, 30]
☐ **B)** [23, 34]
☐ **C)** [11, 23]
☐ **D)** [17, 39]

Mama's answer ▸

Q29 • **MULTIPLE CHOICE** · §1.4 · Apply

Choose ONE

A chip has 10 TFLOP/s peak and 500 GB/s memory bandwidth. Its ridge point is:

- ☐ **A)** 2 FLOP/byte
- ☐ **B)** 20 FLOP/byte
- ☐ **C)** 50 FLOP/byte
- ☐ **D)** 5,000 FLOP/byte

Mama's answer ▸

Q30 • **MULTIPLE CHOICE** · §Lab 1 · Apply · Kitchen Lab

Choose ONE

Kitchen Lab 1: ten 4,096 × 4,096 matrix multiplications (about 687 billion MACs in total) take 0.25 s on a T4 GPU. What throughput did the GPU reach?

- ☐ **A)** 11 TFLOP/s
- ☐ **B)** 0.07 TFLOP/s
- ☐ **C)** 2.7 TFLOP/s
- ☐ **D)** 5.5 TFLOP/s

Mama's answer ▸

Q51 • **OPEN QUESTION** · §1.4 · Apply

A layer has 512 inputs and 2,048 outputs, with FP16 weights, on a chip with 10 TFLOP/s and 500 GB/s. Compute the arithmetic intensity at batch 1 and at batch 32, decide whether each is memory-bound or compute-bound, estimate the achieved TFLOP/s, and explain in your own words why batching helps.

Your answer (more than 100 words, or full working):

Mama's answer ▸

Q52 • **OPEN QUESTION** · §1.5 · Evaluate

Evaluate running Face ID on the phone's Neural Engine instead of in the cloud. Give two advantages and two limitations of on-device AI, and decide whether the same choice would suit a large chatbot model. Justify your decision.

Your answer (more than 100 words, or full working):

Mama's answer ▸



Station 1 scorecard

True/False ____ / 5 **Multiple choice ____ / 5** **Open ____ / 10**

Total ____ / 20 **Stars: ☆ ☆ ☆** **Chef's stamp: (_____)**

STATION 2

Order Up!



Networks & AI infrastructure · 5 T/F · 5 multiple choice · 2 open



Mama's hint for Station 2

Bandwidth is how MUCH, latency is how LONG. Keep them apart and most network questions get easy.

Q6 • **TRUE OR FALSE** · §2.1 · Understand

Encapsulation means each layer adds its own header to the data from the layer above.

☐ **TRUE (Tasty!)** ☐ **FALSE (Burnt!)**

Mama's answer ▸

Q7 • **TRUE OR FALSE** · §2.2 · Understand

UDP guarantees that every packet arrives in order.

☐ **TRUE (Tasty!)** ☐ **FALSE (Burnt!)**

Mama's answer ▸

Q8 • **TRUE OR FALSE** · §2.4 · Understand

Bandwidth and latency are two names for the same quantity.

☐ **TRUE (Tasty!)** ☐ **FALSE (Burnt!)**

Mama's answer ▸

Q9 • **TRUE OR FALSE** · §2.4 · Apply

In a ring all-reduce, the traffic per chip approaches 2 × S as the number of chips grows.

☐ **TRUE (Tasty!)** ☐ **FALSE (Burnt!)**

Mama's answer ▸

Q10 • **TRUE OR FALSE** · §Lab 2 · Apply · Kitchen Lab

In Kitchen Lab 2, a server answers ping with an average round-trip time of 12 ms. Using the speed-of-light floor of 5 µs per km (one way), the server can be at most about 1,200 km away.

☐ **TRUE (Tasty!)** ☐ **FALSE (Burnt!)**

Mama's answer ▸

Q31 • **MULTIPLE CHOICE** · §2.2 · Understand

Select ALL that apply

Which of these does TCP provide that UDP does NOT?

☐ **A)** Acknowledgement of delivered data
☐ **B)** In-order delivery
☐ **C)** Port numbers to reach the right program
☐ **D)** Retransmission of lost packets

Mama's answer ▸

Q32 • **MULTIPLE CHOICE** · §2.2 · Understand

Choose ONE

Which protocol is built on UDP to speed up connection set-up for the web?

☐ **A)** QUIC
☐ **B)** ARP
☐ **C)** SMTP
☐ **D)** FTP

Mama's answer ▸

Q33 • **MULTIPLE CHOICE** · §2.3 · Understand

Choose ONE

A factory camera must stop a machine within 15 ms. The best place to run the model is:

☐ **A)** A distant cloud region
☐ **B)** On or near the machine (edge)
☐ **C)** A user's laptop at home
☐ **D)** Any server; latency does not matter

Mama's answer ▸

Q34 • **MULTIPLE CHOICE** · §2.4 · Understand

Choose ONE

Tail latency refers to:

- ☐ **A)** Upload speed
- ☐ **B)** The average delay
- ☐ **C)** The slowest small fraction of requests
- ☐ **D)** The delay of the last packet only

Mama's answer ▸

Q35 • **MULTIPLE CHOICE** · §2.5 · Understand

Choose ONE

Multi-access edge computing (MEC) in 5G mainly helps AI services by:

- ☐ **A)** Putting computing close to the base station
- ☐ **B)** Replacing TCP
- ☐ **C)** Removing the need for antennas
- ☐ **D)** Increasing screen resolution

Mama's answer ▸

Q53 • **OPEN QUESTION** · §2.3 · Evaluate

A hospital wants an AI that reads X-rays in under 2 seconds and keeps patient data private, but the model needs a large GPU. Recommend cloud, edge (on-premises) or hybrid, and justify the design with latency, bandwidth, privacy and cost.

Your answer (more than 100 words, or full working):

Mama's answer ▸

Q54 • **OPEN QUESTION** · §2.4 · Apply

14 GB of gradients are all-reduced across 256 chips with 100 GB/s per chip, and one compute step takes 8.2 s. Calculate the traffic per chip, the communication time and the scaling efficiency, then explain one way to improve it and why it works.

Your answer (more than 100 words, or full working):

Mama's answer ▸



Station 2 scorecard

True/False ____ / 5 Multiple choice ____ / 5 Open ____ / 10

Total ____ / 20 Stars: ☆ ☆ ☆ Chef's stamp: (_____)

STATION 3

Training the Kitchen Pup

Reinforcement learning · 5 T/F · 5 multiple choice · 2 open





Mama's hint for Station 3

A reward is what you pay, the return is what you collect over time. Watch the γ powers!

Q11 • **TRUE OR FALSE** · §3.2 · Apply

With rewards [0, 0, 0, 10] and γ = 0.9, the discounted return is 7.29.

☐ **TRUE (Tasty!)** ☐ **FALSE (Burnt!)**

Mama's answer ▸

Q12 • **TRUE OR FALSE** · §3.2 · Understand

A discount factor γ close to 0 makes the agent far-sighted.

☐ **TRUE (Tasty!)** ☐ **FALSE (Burnt!)**

Mama's answer ▸

Q13 • **TRUE OR FALSE** · §3.3 · Understand

Reward hacking means the agent earns a high reward without doing the intended task.

☐ **TRUE (Tasty!)** ☐ **FALSE (Burnt!)**

Mama's answer ▸

Q14 • **TRUE OR FALSE** · §3.4 · Understand

A purely greedy agent can get stuck on a poor option forever.

☐ **TRUE (Tasty!)** ☐ **FALSE (Burnt!)**

Mama's answer ▸

Q15 • **TRUE OR FALSE** · §Lab 3 · Understand · Kitchen Lab

In Kitchen Lab 3 (GridWorld), setting epsilon = 0 guarantees that the agent eventually finds the best route.

☐ **TRUE (Tasty!)** ☐ **FALSE (Burnt!)**

Mama's answer ▸

Q36 • **MULTIPLE CHOICE** · §3.2 · Apply

Choose ONE

A corridor of squares 1-2-3 leads to the goal, one step after square 3 (reward 1). With γ = 0.9, V(1) equals:

☐ **A)** 1.0
☐ **B)** 0.729
☐ **C)** 0.81
☐ **D)** 0.9

Mama's answer ▸

Q37 • **MULTIPLE CHOICE** · §3.3 · Apply

Choose ONE

Q = 0.2, r = 1, max Q(s′) = 0.5, α = 0.1, γ = 0.9. After one Q-learning update the new Q is:

☐ **A)** 0.45
☐ **B)** 1.45
☐ **C)** 0.245
☐ **D)** 0.325

Mama's answer ▸

Q38 • **MULTIPLE CHOICE** · §3.3 · Understand

Select ALL that apply

Which of these rewards are DENSE (given at almost every step)?

☐ **A)** +speed every step
☐ **B)** +1 only on reaching the charging station
☐ **C)** −energy used every step
☐ **D)** +1 only for winning the game

Mama's answer ▸

Q39 • **MULTIPLE CHOICE** · §3.4 · Apply

Choose ONE

A bandit has arms that pay 0.2, 0.5 and 0.8 on average. The agent knows the best arm and uses ε = 0.1 (when exploring, it picks one of the three arms at random). Expected reward per pull:

- ☐ **A)** 0.50
- ☐ **B)** 0.77
- ☐ **C)** 0.80
- ☐ **D)** 0.74

Mama's answer ▸

Q40 • **MULTIPLE CHOICE** · §3.5 · Understand

Choose ONE

Domain randomisation helps sim-to-real transfer by:

- ☐ **A)** Removing the reward
- ☐ **B)** Using only real data
- ☐ **C)** Making the simulator perfect
- ☐ **D)** Training on many varied simulated conditions

Mama's answer ▸

Q55 • **OPEN QUESTION** · §3.3 · Create

Design a reward for a home cleaning robot. Show how a naive reward could be hacked, propose a better reward, say whether it is sparse or dense, and compare the two rewards: which behaviour would each one produce?

Your answer (more than 100 words, or full working):

Mama's answer ▸

Q56 ● **OPEN QUESTION** · §Lab 3 · Analyse · Kitchen Lab

A student runs Kitchen Lab 3 (GridWorld TD learning) with epsilon = 0. After 2,000 steps the agent still walks the same long route; the arrows on squares it never visited still show their starting values. Explain why this plan failed, what the student should change, and what they should expect with epsilon = 0.5. Connect the result to the greedy agent in the bandit simulation and compare ε-greedy with UCB as a fix.

Your answer (more than 100 words, or full working):

Mama's answer ▸



Station 3 scorecard

True/False ____ / 5 Multiple choice ____ / 5 Open ____ / 10

Total ____ / 20 Stars: ☆ ☆ ☆ Chef's stamp: (_____)

STATION 4

Chef's Senses

AI sensors · 5 T/F · 5 multiple choice · 2 open





Mama's hint for Station 4

Echo sensors: distance = speed × time ÷ 2. Fusion trusts the more precise sensor more.

Q16 • **TRUE OR FALSE** · §4.1 · Remember

A single camera directly measures the distance to every pixel.

☐ **TRUE (Tasty!)** ☐ **FALSE (Burnt!)**

Mama's answer ▸

Q17 • **TRUE OR FALSE** · §4.1 · Apply

A LiDAR pulse returning after 400 ns means the object is about 60 m away.

☐ **TRUE (Tasty!)** ☐ **FALSE (Burnt!)**

Mama's answer ▸

Q18 • **TRUE OR FALSE** · §4.1 · Understand

Radar measures the speed of an object directly using the Doppler effect.

☐ **TRUE (Tasty!)** ☐ **FALSE (Burnt!)**

Mama's answer ▸

Q19 • **TRUE OR FALSE** · §4.4 · Understand

Two sensors that fail for the same reason (for example fog) give the same benefit as two independent sensors.

☐ **TRUE (Tasty!)** ☐ **FALSE (Burnt!)**

Mama's answer ▸

Q20 • **TRUE OR FALSE** · §Lab 4 · Understand · Kitchen Lab

In Kitchen Lab 4 (phyphox), a phone lying still on a table shows an acceleration (without g) of exactly 0 m/s².

☐ **TRUE (Tasty!)** ☐ **FALSE (Burnt!)**

Mama's answer ▸

Q41 • **MULTIPLE CHOICE** · §4.1 · Understand

Select ALL that apply

Which sensors measure distance DIRECTLY?

☐ **A)** LiDAR
☐ **B)** A single camera
☐ **C)** Radar
☐ **D)** Ultrasonic

Mama's answer ▸

Q42 • **MULTIPLE CHOICE** · §4.2 · Apply

Choose ONE

At 100 km/h, how far does a car travel during 0.3 s of processing latency?

☐ **A)** 8.33 m
☐ **B)** 30 m
☐ **C)** 64.3 m
☐ **D)** 2.78 m

Mama's answer ▸

Q43 • **MULTIPLE CHOICE** · §4.3 · Apply

Choose ONE

Radar says 50.0 m (σ = 0.5 m) and a camera says 52.0 m (σ = 2.0 m). The inverse-variance fused estimate is about:

☐ **A)** 52.0 m
☐ **B)** 50.12 m
☐ **C)** 51.0 m
☐ **D)** 51.5 m

Mama's answer ▸

Q44 • MULTIPLE CHOICE · §4.3 · Apply

Choose ONE

A stereo camera has f = 1,000 px and baseline B = 0.3 m. A disparity of 6 px means a depth of:

- ☐ **A)** 300 m
- ☐ **B)** 42.9 m
- ☐ **C)** 50 m
- ☐ **D)** 60 m

Mama's answer ▸

Q45 • MULTIPLE CHOICE · §4.4 · Understand

Choose ONE

Under SAE J3016, a driver-assist that steers and brakes but needs constant human supervision is:

- ☐ **A)** Level 2
- ☐ **B)** Level 4
- ☐ **C)** Level 5
- ☐ **D)** Level 0

Mama's answer ▸

Q57 • OPEN QUESTION · §4.4 · Apply

A car at 50 km/h has 0.15 s sensing-and-processing latency and brakes at 7 m/s². Compute the total stopping distance and the impact speed if a ball is detected 15 m ahead. What does this show about sensor range and latency when you design a safety system?

Your answer (more than 100 words, or full working):

Mama's answer ▸

Q58 • **OPEN QUESTION** · §Lab 4 · Analyse · Kitchen Lab

In the offline version of Kitchen Lab 4, two students stand 20 m apart. One claps; the other starts a stopwatch on SEEING the clap and stops it on HEARING it. Five trials give t = 0.21, 0.08, 0.15, 0.30 and 0.11 s. Compute the speed of sound from each trial and from the mean time, compare with 343 m/s, and explain why single results scatter so much. Why does averaging help, what can averaging NOT fix, and how would you improve the experiment?

Your answer (more than 100 words, or full working):

Mama's answer ▸



Station 4 scorecard

True/False ____ / 5 Multiple choice ____ / 5 Open ____ / 10

Total ____ / 20 Stars: ☆ ☆ ☆ Chef's stamp: (_____)

STATION 5

Running the Chain

Integration at scale · 5 T/F · 5 multiple choice · 2 open





Mama's hint for Station 5

Big jobs are limited by memory, communication and failures, not just FLOPs.

Q21 • **TRUE OR FALSE** · §5.1 · Apply

Fusing three element-wise operations on a 2 MB tensor cuts memory traffic from 12 MB to 4 MB.

☐ **TRUE (Tasty!)** ☐ **FALSE (Burnt!)**

Mama's answer ▸

Q22 • **TRUE OR FALSE** · §5.2 · Understand

In data parallelism, each chip stores only a slice of the model's weights.

☐ **TRUE (Tasty!)** ☐ **FALSE (Burnt!)**

Mama's answer ▸

Q23 • **TRUE OR FALSE** · §5.4 · Understand

A PUE of 1.0 means the facility uses no energy beyond the computing equipment.

☐ **TRUE (Tasty!)** ☐ **FALSE (Burnt!)**

Mama's answer ▸

Q24 • **TRUE OR FALSE** · §5.4 · Apply

If each of 4,096 chips has an MTBF of 43,800 hours, the whole system fails about every 10.7 hours.

☐ **TRUE (Tasty!)** ☐ **FALSE (Burnt!)**

Mama's answer ▸

Q25 • **TRUE OR FALSE** · §Lab 5 · Understand · Kitchen Lab

In Kitchen Lab 5, the FIRST call of a jax.jit function is much slower than later calls mainly because XLA compiles it.

☐ **TRUE (Tasty!)** ☐ **FALSE (Burnt!)**

Mama's answer ▸

Q46 • **MULTIPLE CHOICE** · §5.2 · Apply

Choose ONE

Training a 7B-parameter model with mixed-precision Adam needs about 16 bytes per parameter. Total training state:

☐ **A)** 14 GB
☐ **B)** 56 GB
☐ **C)** 112 GB
☐ **D)** 224 GB

Mama's answer ▸

Q47 • **MULTIPLE CHOICE** · §5.3 · Apply

Choose ONE

A 7 GB INT8 model is served on a chip with 1 TB/s memory bandwidth. The memory-bound limit for one user is about:

☐ **A)** 1,000 tokens/s
☐ **B)** 4,571 tokens/s
☐ **C)** 14 tokens/s
☐ **D)** 143 tokens/s

Mama's answer ▸

Q48 • **MULTIPLE CHOICE** · §5.5 · Apply

Choose ONE

The diameter (longest shortest path) of a 16 × 16 × 16 torus is:

☐ **A)** 16 hops
☐ **B)** 24 hops
☐ **C)** 45 hops
☐ **D)** 48 hops

Mama's answer ▸

Q49 • **MULTIPLE CHOICE** · §5.4 · Understand

Select ALL that apply

Which changes usually RAISE the MFU (model FLOPs utilisation) of a training job?

- ☐ **A)** Fusing operations to cut memory traffic
- ☐ **B)** Overlapping communication with computation
- ☐ **C)** Making every step wait for a slow data loader
- ☐ **D)** Using a larger batch per chip

Mama's answer ▸

Q50 • **MULTIPLE CHOICE** · §Lab 5 · Apply · Kitchen Lab

Choose ONE

Kitchen Lab 5: you change the matrix size in the JAX program from 4,096 to 8,192. By what factor do the FLOPs of a @ a grow?

- ☐ **A)** 16×
- ☐ **B)** 2×
- ☐ **C)** 4×
- ☐ **D)** 8×

Mama's answer ▸

Q59 • **OPEN QUESTION** · §5.4 · Apply

Estimate the training time, cost and energy for a 7B model on 1.4 trillion tokens with 256 chips (200 TFLOP/s each, 40% MFU, $2 per chip-hour, 400 W per chip, PUE 1.1). Use FLOPs ≈ 6 × parameters × tokens. Which single change would cut all three numbers, and why?

Your answer (more than 100 words, or full working):

Mama's answer ▸

Q60 ● **OPEN QUESTION** · §Lab 5 · Predict · Kitchen Lab

Kitchen Lab 5 runs this JAX program on a Colab T4 GPU (FP32 peak about 8.1 TFLOP/s):

```
f = jax.jit(lambda a: (a @ a).sum())
```

Measured times: size 1,024: first call 0.62 s, second call 0.0021 s; size 4,096: 0.034 s; size 8,192: 0.27 s. Explain the slow first call, compute the TFLOP/s and MFU for each size, and explain why the time ratio from 1,024 to 4,096 is far from the FLOP ratio while 4,096 → 8,192 is close to 8×. Predict the time for size 16,384.

Your answer (more than 100 words, or full working):

Mama's answer ▸



Station 5 scorecard

True/False ____ / 5 Multiple choice ____ / 5 Open ____ / 10

Total ____ / 20 Stars: ☆ ☆ ☆ Chef's stamp: (_____)

Mama's Answer Key

Check each dish, colour your stars, then tap "Back" to return to the question.





Station 1 · The Kitchen Crew

Q1 ✓ TRUE

One multiply plus one add = 2 FLOPs.

◂ Back to the question

Q2 ✓ FALSE

The fix: the CPU is the most flexible chip; the TPU is the least flexible but the most efficient at matrix work.

◂ Back to the question

Q3 ✓ FALSE

The fix: Amdahl's law gives 1 ÷ (0.1 + 0.9/100) ≈ 9.2×; the limit is 10×, set by the serial 10%.

◂ Back to the question

Q4 ✓ TRUE

Arithmetic intensity is about 1 FLOP/byte, far below a ridge point of about 20 FLOP/byte.

◂ Back to the question

Q5 ✓ FALSE

The fix: the speed-up is much SMALLER at size 256 (only a few times), because most GPU cores sit idle and launch overhead dominates; at 4,096 the GPU is roughly 50-100× faster.

◂ Back to the question

Q26 ✓ A) 1.0

$2 \times 1 + (-1) \times 4 + 0.5 \times 6 = 2 - 4 + 3 = 1.0$.

◂ Back to the question

Q27 ✓ C) NPU

An NPU gives the most work per watt within a few watts; GPUs and TPU pods need far more power.

◂ Back to the question

Q28 ✓ D) [17, 39]

$1 \cdot 5 + 2 \cdot 6 = 17$; $3 \cdot 5 + 4 \cdot 6 = 39$.

◂ Back to the question

Q29 ✓ B) 20 FLOP/byte

10 × 10^{12} ÷ (500 × 10^{9}) = 20 FLOP/byte; below 20 the chip is memory-bound.

◂ Back to the question

Q30 ✓ D) 5.5 TFLOP/s

TFLOP/s = 2 × MACs ÷ time = 2 × 6.87 × 10^{11} ÷ 0.25 ≈ 5.5 × 10^{12}. 2.7 forgets the factor 2; 11 doubles it again.

◂ Back to the question

Q51 ✓ 5 marks

Weights 512 × 2,048 = 1,048,576 → 2.1 MB; FLOPs per input 2,097,152. Batch 1 ≈ 1.0 FLOP/byte < 20 → memory-bound, ≈ 0.5 TFLOP/s (5%). Batch 32 ≈ 29.7 FLOP/byte > 20 → compute-bound, ≈ 10 TFLOP/s. Batching reuses each fetched weight for many inputs, raising FLOPs per byte. Marks: intensities (2), classification and TFLOP/s (1), explanation (2). Accept small differences from rounding or from ignoring activation bytes.

◂ Back to the question

Q52 ✓ 5 marks

Advantages: privacy (face data stays in the Secure Enclave), low latency, works offline, energy-efficient NPU. Limitations: limited compute and memory, unsupported operations fall back to the CPU, models must be small or quantised. Large chatbot: usually cloud or hybrid, because its weights exceed phone memory (or a small on-device model + cloud). Marks: 1 per valid point (4), justified decision (1). Accept other valid points, e.g. updates or battery life.

◂ Back to the question



Station 2 · Order Up!

Q6 ✓ TRUE

Like boxes inside boxes: application data → TCP/UDP header → IP header → link header.

◂ Back to the question

Q7 ✓ FALSE

The fix: UDP gives no delivery or ordering guarantee; TCP provides acknowledgements, retransmission and ordering.

◂ Back to the question

Q8 ✓ FALSE

The fix: bandwidth is how much data per second; latency is how long one message takes to arrive.

◂ Back to the question

Q9 ✓ TRUE

Traffic = 2(N − 1)/N × S, which tends to 2S for large N.

◂ Back to the question

Q10 ✓ TRUE

One way = 6 ms = 6,000 µs; 6,000 ÷ 5 = 1,200 km. Real paths are slower, so the server is usually much closer.

◂ Back to the question

Q31 ✓ A, B, D) Acknowledgement of delivered data / In-order delivery / Retransmission of lost packets

TCP adds acknowledgements, ordering and retransmission. Both TCP and UDP use port numbers, so C is wrong.

◂ Back to the question

Q32 ✓ A) QUIC

QUIC (RFC 9000) runs over UDP and combines transport and TLS handshakes.

◂ Back to the question

Q33 ✓ B) On or near the machine (edge)

Only the edge fits a 15 ms budget; a far cloud round trip alone can exceed it.

◂ Back to the question

Q34 ✓ C) The slowest small fraction of requests

For example the 99th-percentile delay; one slow server can delay a whole request.

◂ Back to the question

Q35 ✓ A) Putting computing close to the base station

Operator servers near the base station shorten the round trip.

◂ Back to the question

Q53 ✓ 5 marks

Strong answers choose an on-premises edge GPU server or hybrid: data stays inside the hospital (privacy and law), LAN latency is low, X-ray images are large (bandwidth), the cloud can still be used for training or overflow. Cost: buying a server vs renting. Marks: recommendation (1), four justified factors (4). Accept a well-argued cloud answer that addresses privacy (encryption, anonymisation, contracts) and the 2-second budget.

◂ Back to the question

Q54 ✓ 5 marks

Traffic = 2 × 255/256 × 14 ≈ 27.9 GB; time ≈ 0.28 s; efficiency = 8.2 ÷ (8.2 + 0.28) ≈ 96.7%. Improvements: overlap communication with compute, larger batch per chip (more compute per byte), gradient compression, hierarchical collectives. Marks: 3 calculations, 2 for one improvement with a correct reason.

◂ Back to the question



Station 3 · Training the Kitchen Pup

Q11 ✓ TRUE

$0.9^3 \times 10 = 7.29$.

◂ Back to the question

Q12 ✓ FALSE

The fix: γ close to 0 makes the agent short-sighted (only the next reward counts); γ close to 1 is far-sighted.

◂ Back to the question

Q13 ✓ TRUE

Example: the boat-racing agent circling to hit respawning targets instead of finishing the race.

◂ Back to the question

Q14 ✓ TRUE

It never tries other options, so it never learns that a better one exists (0% best arm in the bandit simulation).

◂ Back to the question

Q15 ✓ FALSE

The fix: with epsilon = 0 the agent never explores; it repeats the first route that worked, some squares are never tried, and it can get stuck on a poor route.

◂ Back to the question

Q36 ✓ C) 0.81

$V(3) = 1$, $V(2) = 0.9 \times 1 = 0.9$, $V(1) = 0.9 \times 0.9 = 0.81$.

◂ Back to the question

Q37 ✓ D) 0.325

Target = $1 + 0.9 \times 0.5 = 1.45$; TD error = 1.25; new Q = $0.2 + 0.1 \times 1.25 = 0.325$.

◂ Back to the question

Q38 ✓ A, C) +speed every step / −energy used every step

A and C give feedback every step. B and D are sparse: they arrive only at the end.

◂ Back to the question

Q39 ✓ B) 0.77

$0.9 \times 0.8 + 0.1 \times (0.2 + 0.5 + 0.8) \div 3 = 0.72 + 0.05 = 0.77$. Careful: if exploration chose only the OTHER two arms the answer would be $0.9 \times 0.8 + 0.1 \times 0.35 = 0.755$.

◂ Back to the question

Q40 ✓ D) Training on many varied simulated conditions

If the policy works across many random variations, the real world looks like one more variation.

◂ Back to the question

Q55 ✓ 5 marks

Naive: +1 per item put in the bin → the robot tips the bin and re-collects. Better: reward the reduction of rubbish on the floor (or clean area), penalties for knocking things over and for energy. Dense if given every step, sparse if only when the room is clean. Comparison: the naive one rewards activity, the better one rewards the real goal. Marks: hack (2), improved reward (2), sparse/dense with comparison (1). Accept other plausible hacks.

◂ Back to the question

Q56 ✓ 5 marks

Failure: epsilon = 0 means no exploration; the agent exploits the first route that reached the goal, so unvisited squares are never updated and a shorter route is never found. Fix: epsilon about 0.1-0.2 (or decaying epsilon). Epsilon = 0.5: values are learned everywhere but the path looks messy and hits the −1 squares more often. Bandit link: the greedy agent found the best arm 0% of the time. ε-greedy explores at random at a fixed rate; UCB adds a bonus $\sqrt{(2 \ln t / n)}$ to little-tried actions and explores where it is uncertain. Marks: diagnosis (2), fix (1), ε = 0.5 prediction (1), comparison (1). Accept other reasonable fixes such as optimistic initial values.

◂ Back to the question



Station 4 · Chef's Senses

Q16 ✓ FALSE

The fix: a camera measures light and colour; depth must be inferred (stereo, motion, learning). LiDAR, radar and ultrasonic measure distance directly.

◂ Back to the question

Q17 ✓ TRUE

$d = c \times t \div 2 = 3 \times 10^{8} \times 4 \times 10^{-7} \div 2 = 60$ m.

◂ Back to the question

Q18 ✓ TRUE

The frequency shift of the echo gives the relative speed in one measurement.

◂ Back to the question

Q19 ✓ FALSE

The fix: correlated failures happen together; in the guide's example both-fail is 5% instead of 0.25% for independent sensors.

◂ Back to the question

Q20 ✓ FALSE

The fix: the reading jitters around 0 (about ±0.05 m/s²) and is never exactly zero; that jitter is sensor noise σ.

◂ Back to the question

Q41 ✓ A, C, D) LiDAR / Radar / Ultrasonic

LiDAR (light time of flight), radar (radio echo) and ultrasonic (sound echo) measure range; a single camera must infer depth.

◂ Back to the question

Q42 ✓ A) 8.33 m

100 km/h = 27.78 m/s; 27.78 × 0.3 ≈ 8.33 m before the car even reacts.

◂ Back to the question

Q43 ✓ B) 50.12 m

Weights $1/\sigma^2$: 4 and 0.25. (4 × 50 + 0.25 × 52) ÷ 4.25 = 213 ÷ 4.25 ≈ 50.12 m: the precise radar dominates.

◂ Back to the question

Q44 ✓ C) 50 m

$Z = f \times B \div d = 1{,}000 \times 0.3 \div 6 = 50$ m.

◂ Back to the question

Q45 ✓ A) Level 2

Level 2 = partial automation; the human remains the driver.

◂ Back to the question

Q57 ✓ 5 marks

13.89 m/s; blind distance 2.08 m; braking $13.89^2 \div 14 \approx 13.78$ m; total ≈ 15.9 m. Detected at 15 m: 12.92 m left to brake, $v = \sqrt{(13.89^2 - 2 \times 7 \times 12.92)} \approx 3.5$ m/s at impact. Lesson: early, reliable detection (range) and low latency decide whether braking works; both must be in the design budget. Marks: 2 + 2 + 1.

◂ Back to the question

Q58 ✓ 5 marks

Speeds: 95, 250, 133, 67, 182 m/s. Mean t = 0.17 s → v ≈ 118 m/s (average the times, not the speeds). True delay is only 20 ÷ 343 ≈ 0.058 s, shorter than human reaction time (about 0.2 s), so reaction errors are as big as the signal. Averaging n trials shrinks random noise by about $\sqrt{n}$ (sensor fusion idea, 4.3) but cannot remove a systematic bias, e.g. reacting faster to sound than to sight. Improvements: a longer distance (100 m → 0.29 s), many more trials, or phyphox Acoustic Stopwatch on two phones (no human reaction). Marks: calculations (2), noise explanation (1), averaging vs bias (1), improvement (1).

◂ Back to the question



Station 5 · Running the Chain

Q21 ✓ TRUE

Unfused: 3 × (read 2 MB + write 2 MB) = 12 MB. Fused: one read + one write = 4 MB.

◂ Back to the question

Q22 ✓ FALSE

The fix: in data parallelism every chip holds a FULL copy of the model and processes a different slice of the batch; splitting the weights is model parallelism.

◂ Back to the question

Q23 ✓ TRUE

PUE = total facility energy ÷ IT energy; 1.0 means zero overhead for cooling and power conversion.

◂ Back to the question

Q24 ✓ TRUE

43,800 ÷ 4,096 ≈ 10.7 h, which is why large jobs need checkpoints.

◂ Back to the question

Q25 ✓ TRUE

The first call traces and compiles the function (about 0.1-1 s); later calls reuse the compiled code and take milliseconds.

◂ Back to the question

Q46 ✓ C) 112 GB

$7 \times 10^9 \times 16$ B = 112 GB: more than one chip's memory, so the state must be split.

◂ Back to the question

Q47 ✓ D) 143 tokens/s

Every token reads all 7 GB: 7 GB ÷ 1 TB/s = 7 ms per token → about 143 tokens/s.

◂ Back to the question

Q48 ✓ B) 24 hops

Torus: 3 × (16 ÷ 2) = 24 hops. 45 is the mesh without wraparound.

◂ Back to the question

Q49 ✓ A, B, D) Fusing operations to cut memory traffic / Overlapping communication with computation / Using a larger batch per chip

A, B and D keep the compute units busier. C leaves the chips idle, so MFU falls.

◂ Back to the question

Q50 ✓ D) 8×

Matrix multiply = $2n^3$ FLOPs; doubling n gives $2^3 = 8\times$. For large sizes the measured time grows close to 8× too.

◂ Back to the question

Q59 ✓ 5 marks

6 × 7 × 10^9 × 1.4 × 10^{12} = 5.88 × 10^{22} FLOPs; 256 × 200 × 10^{12} × 0.4 = 2.05 × 10^{16} FLOP/s → ≈ 2.87 × 10^6 s ≈ 33.2 days; ≈ 204,167 chip-hours ≈ $408,333; 204,167 h × 0.4 kW × 1.1 ≈ 89.8 MWh. Raising MFU (fusion, better batching, overlap) cuts time, cost and energy in proportion. Marks: 4 values, 1 for the change with reason. Accept "more chips" only if the answer notes that cost and energy stay about the same.

◂ Back to the question

Q60 ✓ 5 marks

First call: tracing + XLA compilation. FLOPs = $2n^3$: 1,024 → 2.15 GFLOP ÷ 0.0021 s ≈ 1.0 TFLOP/s (≈ 13% MFU); 4,096 → 137 GFLOP ÷ 0.034 ≈ 4.0 TFLOP/s (≈ 50%); 8,192 → 1.10 TFLOP ÷ 0.27 ≈ 4.1 TFLOP/s (≈ 50%). 1,024 → 4,096 is 64× the FLOPs but only ≈ 16× the time, because at small sizes fixed overhead (launch, transfer) dominates and the GPU is not full. 4,096 → 8,192: 0.27 ÷ 0.034 ≈ 7.9 ≈ 8×, both sizes are compute-bound. Prediction for 16,384: about 8 × 0.27 ≈ 2.2 s, if memory allows. Marks: compilation (1), TFLOP/s and MFU (2), ratio explanation (1), prediction (1). Accept answers that also mention timing noise or GPU clock changes.

◂ Back to the question



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